What Is Beside This Quasar?

An Archival Investigation of J1237+3314

Yashaswy Akella
Independent research project

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Research status. This is an exploratory project report, not a peer-reviewed publication. It documents an analysis of public astronomical data. Independent reproduction of the central measurements is not documented in this release. The project does not establish a new gravitational lens or a confirmed pair of active galactic nuclei.

The result in one minute

Two objects appear very close together in the direction of J1237+3314. One is a previously known quasar; the other is a fainter optical source.

This project reports two main observations:

- A precisely measured compact radio source lies at the position of the fainter optical object.
- A spectrum taken towards that object contains an emission feature. One possible identification would place the line-emitting gas at approximately the same redshift as the bright quasar.

The unresolved question is whether the faint radio source and the quasar are physically associated. The emission could belong to the faint object's host, to gas around the bright quasar, or to another source along the line of sight.

In short: this project documents a potentially useful clue, not a confirmed new astrophysical system.

1 What I was trying to find

The project began as a search for gravitationally lensed quasars using public survey data.

A **quasar** is a very luminous active galactic nucleus, powered by matter accreting onto a supermassive black hole.

A **gravitational lens** can occur when a foreground mass bends the light of a more distant source. A single quasar may then appear as two or more images.

A **dual active galactic nucleus**, or dual AGN, consists of two distinct accreting black holes in a physically associated galaxy system.

These possibilities can look similar in an image. Two nearby points of light can also be unrelated objects that happen to lie along nearly the same direction.

The aim was therefore not simply to find close objects, but to ask:

> Are we seeing multiple images of one source, two physically associated sources, or an unrelated alignment?

2 The objects were already known

J1237+3314 was not discovered in this project. Existing records already contain the bright quasar, the nearby faint detection, the radio source, and candidate listings for the close pair.

The components are labelled as follows:

- **A:** the bright optical quasar, SDSS J123704.05+331423.8, at redshift $z \approx 1.288$.
- **B:** the faint optical source approximately $1.62''$ from A, close to the position of the compact radio source JVAS B1234+335.

An arcsecond is a small angular unit: one degree contains 3600 arcseconds. Angular proximity alone does not establish physical association.

Property	A: bright quasar	B: faint source
Gaia DR3 identifier	1514739204708610432	1514739204711602816
Right ascension, degrees	189.266863	189.266995
Declination, degrees	+33.239947	+33.239511
Gaia G magnitude	17.94	20.86
Redshift of the nucleus	Approximately 1.288	Not independently established

Table 1: Identifiers and approximate positions of the two components. Larger magnitude values indicate fainter optical sources.

The purpose of this work is to examine how the existing measurements relate to the individual components. It does not claim priority for discovering the pair or for every possible interpretation of it.

3 The public data used

The investigation combines several kinds of observations:

Gaia
: Optical positions and astrometric information [1].

Radio catalogues and radio imaging
: The Radio Fundamental Catalog provides a precise radio position. JVAS, FIRST and VLASS provide additional radio measurements and information about radio structure [2, 3, 4, 5].

DESI Legacy Imaging Surveys
: Optical images and catalogue photometry [6].

DESI spectroscopy
: Spectra collected through fibres pointed towards A and B, together with the associated detector and extraction information [7].

These data do not all have the same resolution, aperture, observing date, or calibration. Combining their catalogue entries without checking those differences can produce misleading component identifications.

4 What the analysis reports

The values below are outputs reported by the project's analysis. Their quoted fitting uncertainties do not account for every possible systematic error.

4.1 The compact radio source is located at B

The radio position differs from B's Gaia position by approximately 4.7 milliarcseconds, or 0.0047″. The same radio position is approximately 1.619″ from A.

The reported offset is consistent with the combined positional uncertainties.

This supports associating the compact radio emission with the position of B rather than A. It does not, by itself, establish B's distance or prove that it is a second nucleus physically associated with A.

4.2 A spectral feature is present towards B

The analysis finds an emission feature near 8529 Å in the spectrum collected through the fibre centred on B.

An emission feature is a narrow wavelength range in which a source produces extra light. Identifying its atomic origin can provide a redshift, but a single feature may have more than one possible identification.

Measurement	Reported value
Feature wavelength	8528.6 ± 0.4 Å
Integrated feature flux	$(2.78 \pm 0.16) \times 10^{-16}$ erg s^{-1} cm^{-2}
Intrinsic width under a single-Gaussian model	550 ± 30 km s^{-1}, FWHM
Gas redshift if identified as [O II]	$z \approx 1.2872$–1.2879

Table 2: Main spectral measurements reported by the project. Wavelengths are reported in the vacuum, heliocentric convention. The flux follows the DESI point-source calibration; it is not simply the uncorrected flux entering the fibre. Uncertainties are statistical fitting estimates.

The feature is reported in both individual exposures and in an additional extraction from the preprocessed detector images. It is also present before subtracting the estimated contribution from A.

These are useful internal checks. However, the exposures use the same fibre and instrument, so they do not eliminate every possible instrumental effect.

4.3 Why the possible oxygen identification matters

One plausible identification is [O II], emission from singly ionized oxygen.

If that identification is correct, the gas has a redshift close to A's redshift. Redshift describes how much wavelengths have shifted relative to their emitted values:

$$z = \frac{\lambda_{\text{observed}}}{\lambda_{\text{emitted}}} - 1.$$

Here, the proposed oxygen feature is a blended doublet. Its components are not separately established by the observed profile, and the inferred redshift depends partly on their unknown relative strengths.

The partner-line tests provide evidence against some alternative identifications, but other possibilities remain viable. Consequently, this report describes the feature as **consistent with** [O II], not as a uniquely identified line.

4.4 Light from A contaminates B's spectrum

A spectroscopic fibre does not collect light from one mathematical point. Atmospheric blurring can send light from a nearby object into the fibre.

The spectrum taken towards B contains broad quasar features that the analysis attributes mainly to light from A. This complicates any attempt to classify B using the spectrum as a whole.

The narrow feature near 8529 Å persists over the tested range of contamination corrections. The spectrum centred on A shows no corresponding significant feature in the forced-line measurement.

However, A and B were observed at different dates and under different seeing conditions. Their comparison is therefore not a simultaneous, spatially resolved measurement of both components.

4.5 The radio constraint at A is model-dependent

The 2023 VLASS fit assigns approximately 144 mJy at 3 GHz to B. No separate component is securely detected at A.

The adopted fitting procedures give sensitivity thresholds of roughly 7–10 mJy at A, but residual structure around the bright radio source limits the interpretation.

These thresholds are not calibrated, model-independent confidence limits. They do not establish that A is radio-quiet.

5 What these observations do not establish

5.1 They do not confirm a dual AGN

Even if the feature is [O II] at A's redshift, the gas might not belong to the compact radio nucleus.

Possible explanations include:

- Gas in B's host galaxy, physically associated with A.
- Gas around A that extends towards B.

- Gas associated with an interaction between galaxies.
- An unrelated radio source projected onto line-emitting gas near A.

A compact radio feature associated with a jet from A is another alternative discussed in the technical analysis.

If B is a distinct active nucleus at A's redshift, the angular separation would correspond to approximately 14 kpc in projection under the cosmology adopted in the technical report. That distance is conditional on physical association; it is not a measured three-dimensional separation.

5.2 They do not definitively exclude gravitational lensing

The optical and radio brightness patterns do not resemble a simple lens model with the same image flux ratio at every wavelength.

Nevertheless, dust, microlensing, different emitting-region sizes, and aperture effects can alter observed ratios. The radio constraint at A is also model-dependent.

The appropriate conclusion is therefore:

> A straightforward two-image lens does not provide an obvious explanation of all the observations, but this project does not establish a definitive rejection of lensing.

5.3 They do not establish a first discovery

The sources and pair were already catalogued. Whether the particular emission feature or its interpretation has previously been reported has not been fully established here.

The potential contribution is a documented reanalysis and a testable observational clue, rather than the discovery of the objects themselves.

6 Limitations and the most useful next test

The main limitations are:

1. The central measurements have not yet been independently reproduced by another researcher.
2. The emission-line identification is not unique.
3. A spectrum collected through a seeing-blurred fibre does not precisely locate the line-emitting gas.
4. The two optical components and their radio emission are not cleanly resolved in every dataset.
5. The radio thresholds and spatial constraints depend on adopted models and assumptions.

The highest-value next step is an independent examination of the DESI feature and radio–optical association.

If further observations become available, spatially resolved spectroscopy could test whether the emission peaks on B and measure additional lines. Higher-resolution radio imaging could separate the radio contributions and better constrain the source structure.

These tests answer different questions. Neither should be described as automatically confirming a dual AGN.

7 How to inspect or reproduce the analysis

The repository should provide the matching code, numerical outputs, data references, and software requirements for one consistent release.

The principal scripts are:

- `astrometry.py`: catalogue-position comparison.
- `desi_extract_coadd_rows.py`: selection of the relevant DESI targets and spectral arrays.
- `desi_feature_fit.py`: feature measurements.
- `desi_line_identification.py`: partner-line tests.
- `detector_trace_check.py`: detector-level comparison.
- `radio_profile.py`: model-dependent radio fits.
- `localization.py`: conditional aperture modelling.

Use the repository README for the authoritative execution instructions. The supplied release documents the following workflow from its scripts directory:

```
cd release/scripts
./fetch_inputs.sh
PYTHON=python3 ./run_all.sh ../manuscripts
```

The provenance document identifies the input datasets and file checksums. The generated results table connects reported numbers to their producing scripts and output fields.

A manuscript-number checker can help find inconsistencies. It does not verify the physical interpretation, demonstrate correct uncertainty coverage, or replace examination of the original data.

Repeated runs of the same code demonstrate repeatability. Independent reproduction is a separate step.

8 Why share an unresolved investigation?

An investigation need not end with a confirmed discovery to be worth documenting.

This project records a practical attempt to combine catalogue positions, optical images, radio observations and fibre spectra. It also illustrates why close-source classifications can remain uncertain even when many public measurements are available.

The purpose of sharing the repository is to make the reasoning and analysis available for inspection, correction, and possible reuse.

Bottom line. A known close pair has been re-examined using public data. The analysis supports locating compact radio emission at the faint optical component and reports a spectral feature that may be relevant to its association with the bright quasar. The physical interpretation remains unresolved.

Data credits and AI assistance

This project uses public data and services from Gaia, DESI, the DESI Legacy Imaging Surveys, SDSS, JVAS, FIRST, VLASS, the Radio Fundamental Catalog, SIMBAD and VizieR. Additional supporting analyses use WISE and unWISE.

Survey data remain subject to their providers' terms and acknowledgement requirements. This project does not claim ownership of the original observations. The repository should accurately record any other tools used and the checks actually performed by the author.

References

[1] Gaia Collaboration et al. (2023). Gaia Data Release 3: summary of the content and survey properties. *Astronomy & Astrophysics*, 674, A1.

[2] Wilkinson, P. N., Browne, I. W. A., Patnaik, A. R., Wrobel, J. M., and Sorathia, B. (1998). Interferometer phase calibration sources—III. *Monthly Notices of the Royal Astronomical Society*, 300, 790–816.

[3] Becker, R. H., White, R. L., and Helfand, D. J. (1995). The FIRST survey: Faint Images of the Radio Sky at Twenty Centimeters. *The Astrophysical Journal*, 450, 559.

[4] Lacy, M. et al. (2020). The Karl G. Jansky Very Large Array Sky Survey: science case and survey design. *Publications of the Astronomical Society of the Pacific*, 132, 035001.

[5] Petrov, L. and Kovalev, Y. Y. (2025). Radio Fundamental Catalog. *The Astrophysical Journal Supplement Series*, 276, 38.

[6] Dey, A. et al. (2019). Overview of the DESI Legacy Imaging Surveys. *The Astronomical Journal*, 157, 168.

[7] Guy, J. et al. (2023). The spectroscopic data processing pipeline for the Dark Energy Spectroscopic Instrument. *The Astronomical Journal*, 165, 144.